

CE 310 — Week 7 Weekly Quiz

Week: 7 | **Topic:** Correlation, Covariance, and Scatter Plots

Due: Closes **8:00 AM Friday, Oct 9** — brought forward because Week 8 is midterm week. Taken on D2L

Points: 15 questions $\times$ 1 pt each = 15 pts

Attempts: Two — the higher score counts

Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Excel tutorial — the Pearson correlation coefficient, covariance, scatter plots, and the Excel functions that go with them. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. The Pearson correlation coefficient r between two variables always falls in which range?

- A) 0 to 1
 - B) -100 to +100
 - C) -1 to +1
 - D) 0 to ∞
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Q2. If $r = -0.85$ between two variables X and Y , what does this mean?

- A) As X increases, Y tends to decrease — a strong negative linear relationship
 - B) As X increases, Y also increases — a strong positive linear relationship
 - C) X causes Y to decrease, because the two variables move in opposite directions
 - D) The variables have no relationship with each other
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Q3. If $r = 0$ between outdoor temperature and cooling energy for a particular dataset, the best interpretation is:

- A) Outdoor temperature definitely does not affect cooling energy in this building
 - B) Both variables must have a mean of zero, so no relationship of any kind can exist between them in this sample
 - C) The dataset has fewer than 100 rows
 - D) There is no *linear* relationship between the two variables — a non-linear relationship could still exist
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Q4. [True/False] A Pearson correlation of $r = 0.9$ between two variables is strong enough to establish that one of them causes the other.

Q5. [Select all that apply] Which of the following are true of the Pearson correlation coefficient r ?

- (A) It is dimensionless, so it can be compared across variable pairs with different units
 - (B) It detects any relationship between two variables, linear or not
 - (C) Its sign gives the direction of the linear relationship
 - (D) Squaring it gives the fraction of variance linearly explained
 - (E) A value near ± 1 establishes that one variable drives the other
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Q6. Which statement correctly distinguishes covariance from correlation?

- A) Covariance is always larger than correlation, no matter what units the two variables happen to be measured in
 - B) Covariance and correlation always give the same numerical value
 - C) Correlation measures how two variables move together; covariance does not
 - D) Covariance has units (e.g., $^{\circ}\text{F} \times \text{kWh}$); correlation is dimensionless and always falls between -1 and $+1$
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Q7. The coefficient of determination r^2 tells you:

- A) The slope of the regression line between two variables
 - B) The fraction of variance in one variable linearly explained by the other
 - C) The probability that two variables are related
 - D) The square root of the covariance between two variables
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Q8. In a machine learning project, you compute all pairwise Pearson r values between your input features. You find that Feature A and Feature B have $r = 0.97$. The main concern is:

- A) Both features will improve the regression model equally

- B) The features are negatively correlated, which makes regression unstable
- C) Both carry nearly identical information, so including both creates multicollinearity
- D) You should remove both features because $r > 0.9$ means they are noise

Q9. [Matching] Match each Excel function to what it returns for two columns x and y.

Choices:

- (A) =CORREL(x, y)
- (B) =COVARIANCE.S(x, y)
- (C) =CORREL(FILTER(x, season="Summer"), FILTER(y, season="Summer"))
- (D) =COVARIANCE.S(x, y) / (STDEV(x) * STDEV(y))
- (E) =STDEV(y)

Matches:

1. The same association measured inside one subgroup only
2. A dimensionless number between -1 and +1
3. The identity that converts the units-carrying measure into the dimensionless one — it should equal CORREL
4. A measure of joint variation carrying the units of $x \times y$

Q10. [Ordering] You suspect a pooled correlation is hiding something. Put the steps of a conditional-correlation check in the order you would carry them out.

- (A) Compute r over the full dataset
- (B) Split the rows into meaningful subgroups — summer and winter, say, or occupied and unoccupied
- (C) Compute r again separately within each subgroup
- (D) Compare the subgroup values against the pooled one, and report the pooled figure as misleading if the sign reverses

Q11. You find that r between zone temperature and cooling energy is weak, although the two are physically linked. What is the most likely explanation?

- A) Zone temperature is in °F and cooling in kWh, and correlation needs matching units
- B) Zone temperature is held near setpoint, so $\sigma \approx 2.9$ °F and it barely varies
- C) The two columns cover different numbers of rows, so they cannot be compared

- D) Correlation only works when both variables are right-skewed, and these are not
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Q12. =CORREL(B2:B8761, C2:C8761) — what happens if you point the second range at C2:C8760 by mistake?

- A) Excel returns #N/A, because CORREL needs both ranges the same length
 - B) Excel pads the shorter range with zeros and returns a slightly biased r
 - C) Excel returns a correlation computed on the overlapping rows only, silently
 - D) Excel returns #DIV/0!, because one range ends before the other does
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Q13. You compute $r = -0.87$ and then r^2 . What is r^2 , and what has changed?

- A) $r^2 = 0.76$, and the sign is gone — it reports variance share, not direction
 - B) $r^2 = -0.76$, keeping the sign so the direction of the relationship survives
 - C) $r^2 = 0.93$, the square root of r rather than its square
 - D) $r^2 = 0.87$, because squaring does not change a negative correlation
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Q14. =COVARIANCE.S(B2:B501, C2:C501) returns 412.6 and =CORREL(...) on the same columns returns 0.83. Why report the correlation rather than the covariance?

- A) Covariance carries the units of both variables, so its size means little alone
 - B) Covariance is only valid for samples of more than 500 rows, and this one qualifies
 - C) Covariance cannot be negative, so it hides the direction of the relationship
 - D) They measure different relationships; covariance is non-linear and r is linear
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Q15. This data calls for a scatter chart with a trendline rather than a line chart. Why does the distinction matter here?

- A) A line chart is not available for numeric x-values in Excel, so scatter is forced
- B) A scatter plots each row as an (x, y) point; a line chart joins them in order

- C) A trendline can only be added to a scatter chart, and never to a line chart
- D) Scatter charts recalculate r automatically where line charts do not